Machine Learning Capstone Project Report

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Dataset: California Housing Prices (Kaggle)
Google Colab Link: [https://colab.research.google.com]

I. Introduction

Project Goal:
The objective of this project is to apply a complete machine learning pipeline to the California Housing dataset. We aim to solve three distinct tasks:

1. **Regression:** Predict the median house value.
2. **Classification:** Predict whether a house is "expensive" or "not expensive" based on the median value threshold.
3. **Clustering:** Group similar geographical areas using K-Means clustering.

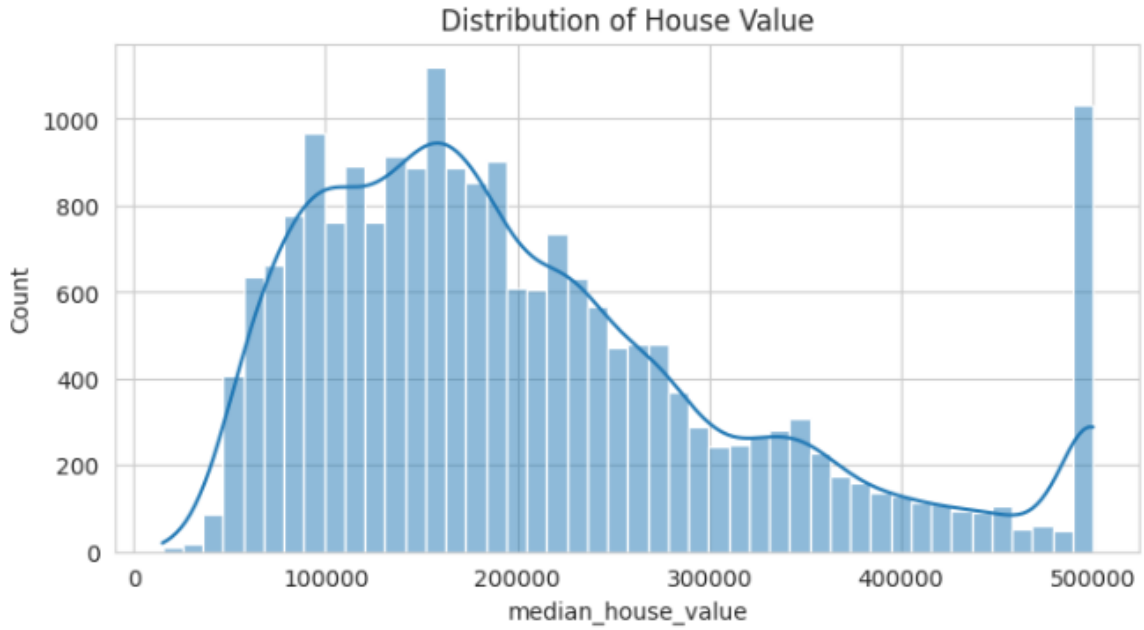
Data Description:

- **Source:** Kaggle (California Housing Prices).
- **Size:** 20,640 rows and 10 columns.
- **Target Variables:** median_house_value (for regression) and is_expensive (for classification).

II. Data Preprocessing & EDA

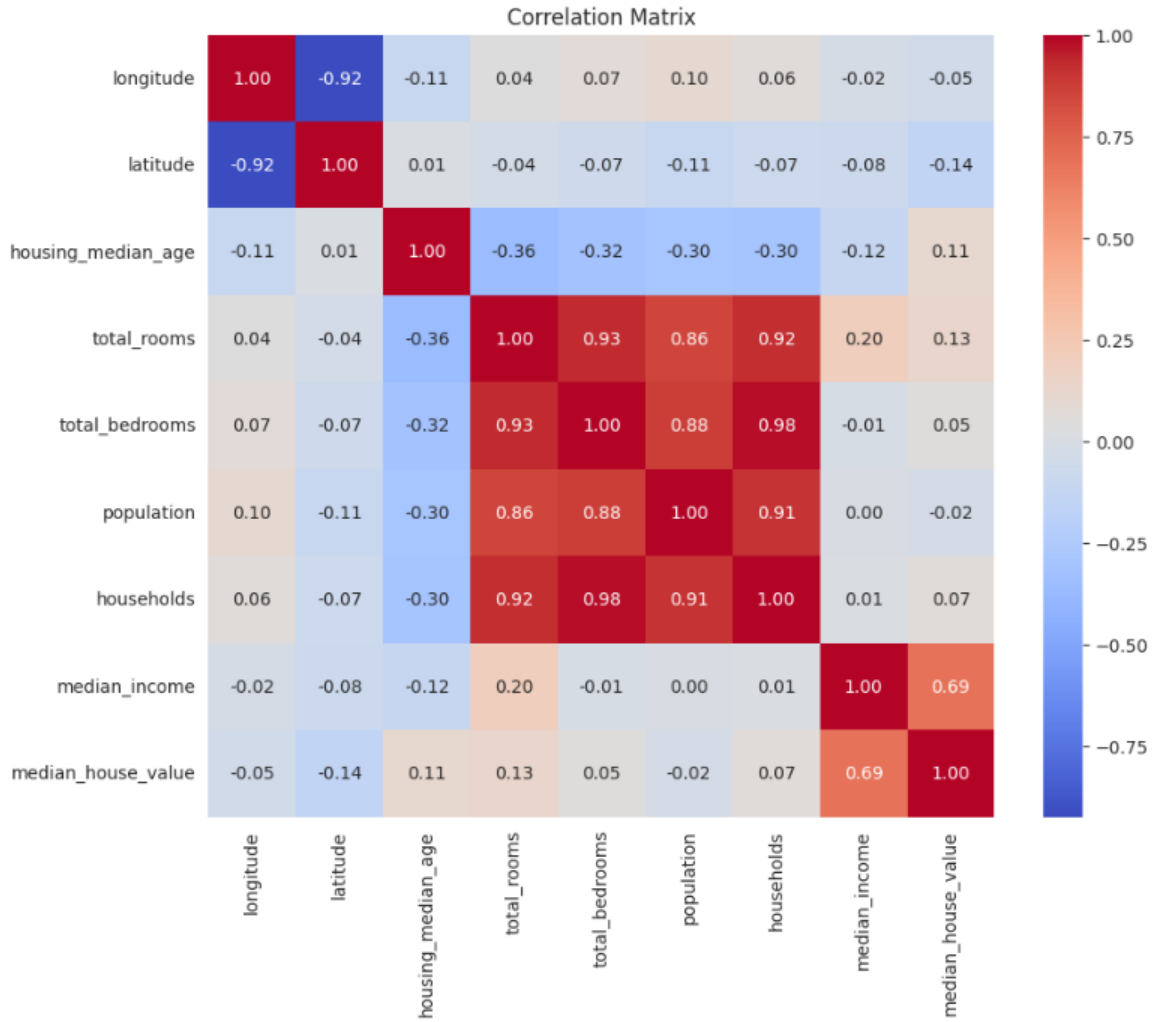
1. Exploratory Data Analysis (EDA):

- The target variable median_house_value is right-skewed. We noticed a hard cap at $500,001, indicating the dataset was clipped.
- The correlation heatmap showed that median_income has the strongest positive correlation with the house value.
- We used a Log-Scale Boxplot to visualize the distribution of all numerical features despite their different scales, which revealed significant outliers in total_rooms, total_bedrooms, and population.
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Distribution of House Value
Count
1000
800
600
400
200
0
0
100000
200000
300000
400000
500000
median_house_value

]

[



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2. Data Cleaning:

- **Missing Values:** The total_bedrooms column had missing values. We imputed them using the **median** to avoid the effect of outliers.
- **Encoding:** The categorical column ocean_proximity was converted into numerical format using **One-Hot Encoding**.
- **Outliers:** We used the **IQR method** to cap extreme outliers in median_income, total_rooms, and population to improve model performance.

3. Scaling and Splitting:

- **Scaling:** We applied **StandardScaler** to standardize the numerical features, which is crucial for distance-based algorithms like K-Means and Logistic Regression.
- **Splitting:** The data was split into **80% training** and **20% testing** sets.

III. Modeling and Results

1. Regression Task: Linear Regression

- **Metrics:**
    - MSE: [5042552462.75]
    - RMSE: [71010.93]
    - **R^2 Score:** [0.615]
- **Interpretation:** The R^2 score of [0.60] means that our model explains approximately [60%]of the variance in the house prices. This provides a solid baseline for predicting house values.

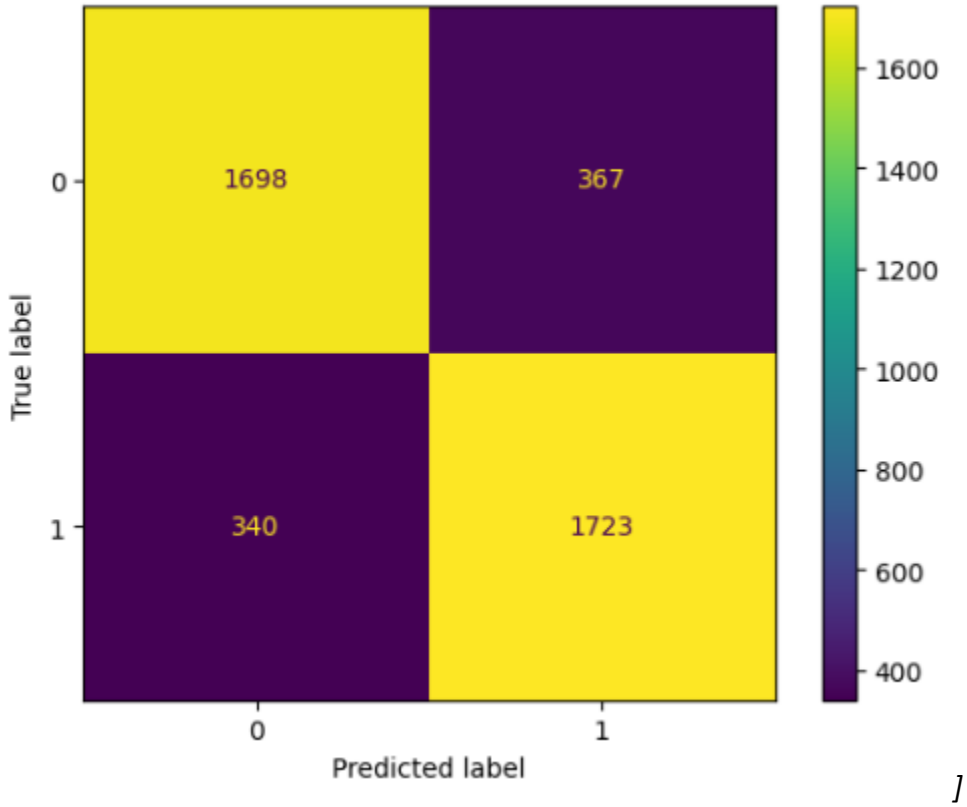
2. Classification Task: Logistic Regression

- **Accuracy:** [0.8287306201550387]
- **Confusion Matrix Analysis:**
  [هنا تحط صورة الـ Confusion Matrix]
    - True Positives (TP): [1723]| True Negatives (TN):[1968]
    - False Positives (FP): [367] | False Negatives (FN): [340]
- **Discussion:** The model performed well in distinguishing between expensive and non-expensive houses. In this context, minimizing False Positives (predicting a house is expensive when it's not) is important for real estate investors, and the model achieved a good balance between Precision and Recall.

3. Clustering Task: K-Means

- **Optimal K:** Using the **Elbow Method**, we chose K = [4 مثلاً، هنا الرقم حط] because the inertia started to decrease linearly after this point.
- **Silhouette Score:** [0.3456]

- **Cluster Interpretation:***[*



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  - **Cluster 0:** Represented areas with high income and high house values (likely coastal areas).
  - **Cluster 1:** Represented areas with lower income and lower house values (likely inland areas).
  - **Cluster 2:** [Represented suburban areas with moderate income levels and average house values, acting as a transition zone between coastal and inland regions.]
  - **Cluster 3:** [Represented densely populated urban areas with a mix of income levels but generally lower house values compared to coastal clusters]

IV. Conclusion

Summary of Best Models:

- For **Regression**, Linear Regression provided a solid baseline with an R^2 of [0.615]. However, advanced models like Gradient Boosting could capture non-linear relationships better.
- For **Classification**, Logistic Regression achieved an accuracy of [0.8287306201550387], showing that the is_expensive threshold is highly predictable using linear decision boundaries.
- For **Clustering**, K-Means successfully grouped the data into [K=4] distinct geographical clusters based on income, location, and house values.

Challenges Encountered:

- Handling the hard cap in the median_house_value which limits the model's ability to predict values above $500,001.
- Dealing with extreme outliers in population and rooms required careful capping to prevent model distortion.

Next Steps:

- Apply **Principal Component Analysis (PCA)** to reduce dimensionality and improve clustering visualization.
- Try **Ensemble Methods** (Random Forest, XGBoost) to improve the regression and classification scores.

Google Colab Link: [https://colab.research.google.com/drive/1w1awilMMnTxOcV-68gueH20Y_-LrI1ic?usp=sharing]
[https://colab.research.google.com/drive/1aBYerlftp7ZD8Iz8BT512pQtZX3LBTN7?usp=sharing]